GlycoTwin

Type 2 diabetes twin. Predict whether glucose reaches 180 mg/dL in the next 2 hours.

Add the team name and college on this page before submission.

Patients: 339. Decision rows: 222384.
Diabetes 2-hour spike rate: 0.3707.
Prediabetes 2-hour spike rate: 0.0557.
Split by patient, seed 2026: 70% train, 15% validation, 15% test.

Data

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Static record: Synthea sample, diabetes and prediabetes only.
HbA1c below a diabetic range is kept as hba1c_raw and replaced for the
simulator. The source column says synthea_lab or imputed_from_diagnosis.

Wearables: 14 days, every 5 minutes, generated with seed 2026 from
28.7 x HbA1c - 46.7, meals at 08:00, 13:30 and 21:00, steps, sleep, and insulin.
Sleep is one value per night, copied onto that calendar day.

Rebuild: py -3 data/generate_signals.py
         py -3 backend/src/make_dataset.py
Check:   py -3 backend/src/validate_data.py
```

Held-out result

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LightGBM AUROC: 0.9857
Current glucose alone AUROC: 0.7505
30-minute slope alone AUROC: 0.6191
False alarms per patient-day at probability 0.60: 0.272
Curve mean absolute error (mg/dL): {'30': 5.46, '60': 5.91, '90': 6.36, '120': 7.55}

Same current glucose, 140 to 160 mg/dL:
  spike rate if HbA1c < 7: 0.0083
  spike rate if HbA1c >= 7: 0.2972

The score is on simulated patients. It is not a clinical validation.
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